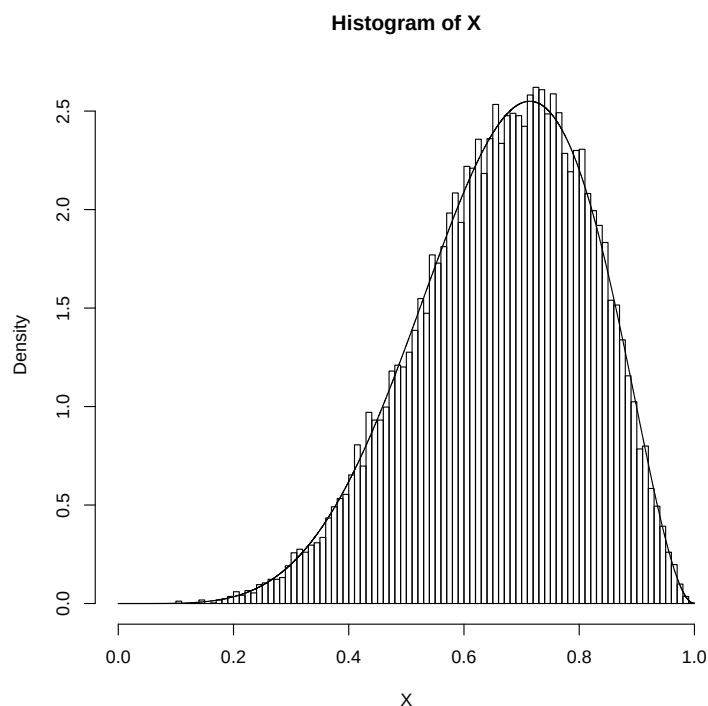


Homework for Chapter 3

1. Generate a random sample of size 1000 from the Beta(3,2) distribution using the acceptance-rejection method. Graph the histogram of the sample with the theoretical Beta(3,2) density superimposed as the Figure below.



2. Generate a random sample of size 1000 from the pdf

$$f(x) = \frac{2}{\pi R^2} \sqrt{R^2 - x^2}, \quad -R \leq x \leq R$$

using the acceptance-rejection method. Graph the histogram of the sample with the theoretical density superimposed.

3. The continuous random variable X with positive support is said to have the Pareto distribution if its probability density function is given by

$$f(x) = \frac{\beta \alpha^\beta}{(x + \alpha)^{\beta+1}}, \quad x > 0.$$

Generate a random sample of size 1000 from the Pareto distribution with $\alpha = 2$ and $\beta = 4$ using "inverse transformation method". Compare the empirical and theoretical distributions by graphing the histogram of the sample and superimposing the Pareto density curve.

4. Please compare the efficiencies of two methods for generating the Wishart Distribution with $n = 100$ and covariance $\Sigma = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 2 \end{pmatrix}$.
5. Please generate data from the AFT (Accelerated Failure Time) model to evaluate how much loss of power the traditional Cox model would produce when the proportionality risk assumption is violated. Run a 1,000-iteration Monte Carlo power simulation and plot comparative power curves across 30%, 50%, and 70% censoring rates. For a baseline Weibull distribution with scale λ and shape ν , the baseline survival function is $S_0(t) = \exp(-\lambda t^\nu)$. Substituting the accelerated time into $S_0(t)$ gives:

$$S(t|X) = \exp(-\lambda (t \cdot \exp(X\beta))^\nu)$$

Using the Inverse Probability Transform Method to generate survival time T .